

Ion selectivity

Two buffers were used: Buffer A—2 M KCl, 15 mM HEPES, pH 7.5; Buffer B—0.5 M KCl, 15 mM HEPES, pH 7.5; Buffer C—0 M KCl, 15 mM HEPES, pH 7.5; Buffer D—2 M KCl, 50 mM citric acid, BTP, pH 3.8; Buffer E—0.5 M KCl, 50 mM citric acid, BTP, pH 3.8; and Buffer F—0 M KCl, 50 mM citric acid, BTP, pH 3.8.

The ion selectivity of the CytK mutants was determined in buffers of either pH 7.5 or pH 3.8. First, both the *cis* and *trans* compartments were filled with 2 M buffer (Buffer A or Buffer D), and a single nanopore was isolated, followed by the pipet offset adjustment of the current at 0-mV bias to 0 pA. The I/V curve was determined between −140 mV and +140 mV, in steps of 20 mV, using a 10-kHz sampling rate coupled with a 2-kHz Bessel filter. Next, the concentration of KCl in the *trans* compartment was lowered to approximately 0.5 M by flushing with 0 M buffer (Buffer C/Buffer F), and repeated flushing with 0.5 M buffer (Buffer B/Buffer E) ensured that the final concentration of KCl in the *trans* compartment was correctly fine-tuned to 0.5 M. Similarly, the I/V curve of the nanopore was measured. The reversal potential was determined from the second I/V curve from the linear function fitting the data points between −20 mV and +20 mV. The ion selectivity, expressed as the fraction p_{K^+}/p_{Cl^-} , was calculated using the formula below, and each ion selectivity was established from triplicate experiments.

$$\frac{p_{K^+}}{p_{Cl^-}} = \frac{[a_{Cl^-}]_{trans} - [a_{Cl^-}]_{cis} \times e^{V_r F/RT}}{[a_{K^+}]_{trans} \times e^{V_r F/RT} - [a_{K^+}]_{cis}}$$

where $[a]$ is the activity of the K^+ or Cl^- in the *cis* or *trans* compartment; V_r is the reversal potential, which is obtained from the experiments; F corresponds to the Faraday constant ($96,485 \text{ C mol}^{-1}$); R is the gas constant ($8.3145 \text{ J mol}^{-1} \text{ K}^{-1}$); and T is the temperature (298 K).